"Osprey 3.5 V6: Low fuel rail pressure from failing high-pressure fuel pump"

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1 Condition

Osprey 3.5 V6 vehicles may set diagnostic trouble code P0087 (fuel rail/system pressure too low) and may also store P0190 (fuel rail pressure sensor circuit). Drivers may report extended cranking, a stumble or hesitation under acceleration, and an illuminated malfunction indicator lamp. P0087 reflects measured rail pressure falling below the commanded target during demand. Begin with the Osprey Fuel Service Manual :: Fuel System Diagnosis .

2 Cause

The condition is most often caused by internal wear of the high-pressure fuel pump HPFP-4120, which reduces volumetric output and prevents the rail from reaching commanded pressure, setting P0087. A worn pump follower or degraded internal seal lowers peak delivery, particularly under high load. Rule out a restricted supply or sensor bias before condemning the pump. Pump operation and specifications are covered in the Osprey Fuel Service Manual :: High-Pressure Fuel Pump (GDI) .

3 Repair Procedure

Run PROC-FUEL-DIAG to capture commanded versus actual rail pressure and confirm the deficit originates at the pump rather than the sensing circuit. Replace the high-pressure fuel pump HPFP-4120 if output is below specification, and verify the integrity of the fuel rail pressure sensor FRS-4180 since a biased sensor can mimic or mask P0087. The structured pressure and fuel trim test is documented in the Osprey Fuel Service Manual :: Fuel Trim and Pressure Diagnosis Procedure , and FRS-4180 service is in the Osprey Fuel Service Manual :: Fuel Rail Pressure Sensor Replacement . For interpretation of related pressure and sensor codes refer to the Osprey Fuel Service Manual :: Fuel Pressure and Sensor Faults . After repair, clear P0087 and P0190 and confirm rail pressure tracks the target through a road test.